
Design and implementation of 31 level asymmetrical cascaded MLI with DC-DC flyback converter for photovoltaic system using P&O technique for electric traction application

J. Gowri Shankar and J. Belwin Edward*

School of Electrical Engineering,
VIT University,
Vellore, 632014, India
Email: jananishankar18@gmail.com
Email: jbelwinedward@vit.ac.in

*Corresponding author

Abstract: This paper proposes a design and implementation of 31-Level MLI (Multi-Level Inverter) which is supplied from a single solar panel with multi-output flyback converter for transformer-less railway traction drive. The multi-output flyback converter provides isolation for the circuit and also produces four different supply voltages based on the transformer turns ratio. In asymmetrical MLI the DC source magnitudes are unequal and it is designed with binary form of voltage such as 15Vdc, 30Vdc, 60Vdc and 120Vdc. In this proposed technique, we are using a single solar panel and single fly back converter, from this four DC output voltages are obtained in the ratio of 1 : 2 : 4 : 8. Our proposed technique requires fewer switches, i.e., seven switches for more levels, i.e., 31 levels, and it also gives less total harmonic distortion (THD). The results are verified in MATLAB, PROTEUS and real-time system using TMS320F2802 and ATMEGA328.

Keywords: PV system; flyback converter; asymmetrical cascaded MLI; TMS320F2802.

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Biographical notes: J. Gowri Shankar received his BE in Electrical Engineering from Anna University, Tamil Nadu, India in 2005, an MTech in Power Electronics and Drives from VIT University, Tamil Nadu, India in 2008. From 2015, he is pursuing PhD in VIT University, Vellore, in the area of Asymmetrical Cascaded Multilevel inverter for Solar Photovoltaic system. He has more than 10 years of experience in teaching and research. His major scientific interest is focused on multilevel inverter, renewable energy sources.

J. Belwin Edward received his BE in Electrical Engineering from Manonmaniam Sundaranar University, Tamilnadu, India in 1999, an ME in Power Systems from Annamalai University, Tamilnadu, India in 2002 and a PhD from Anna University, Chennai in 2013. He has more than 16 years of experience in teaching and research. His major scientific interest is focused on power system optimisation and protection, optimal location of FACTS and DG devices, renewable energy systems.

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1 Introduction

In recent years, multi-level inverters are more popular in electric high-power applications with fewer conflicts and the possibility to function at lower switching frequencies compared to regular two-level inverters. The multi-level inverter (MLI) yields common mode voltage, dropping the pressure of the motor without affecting the motor. With low distortion, it is possible to draw the input current by MLI. The MLI can function at both frequencies of high switching and low switching. In case of lower frequency, it has fewer losses and it achieves higher efficiency. Discriminating harmonic removal method through the multi-level topography outcomes, the total harmonic distortion (THD) develops less in the production waveform beyond consuming at the circuit of all filter. Basically, MLIs are available in three classes: neutral point diode clamped (NPDCMLI), flying capacitor (FCMLI) and cascaded H-bridge (CHBMLI). The block diagram of proposed 31-level MLI as shown in Figure 1. Neutral point clamped MLI comprises clamping diodes and it increases the levels of voltage. In order to balance the voltage, the capacitor is associated in series (Nami et al., 2011). This makes an immense problem for strategies. But in case of flying capacitor MLI, more clamping capacitors are connected, therefore it is difficult to balance the voltage (Han et al., 2014). Finally, the cascaded H-bridge MLI is applicable for high voltage applications since, every H-bridge contains of four switches and one DC source. In this, depending upon DC source, it is divided into two types:

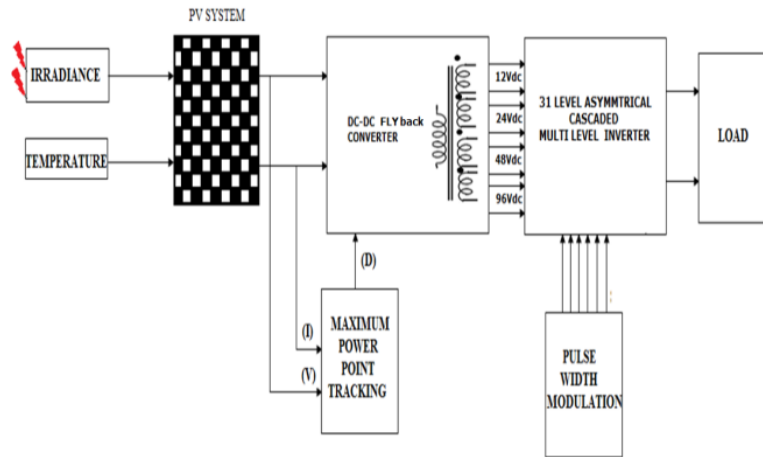
- single DC source
- multiple DC source (Nami et al., 2011).

The multi-level voltage source inverters’ unique structure allows them to reach high voltages and power levels without the use of transformers (Shahir and Babaei, 2016). They are especially suited to high voltage vehicle drives where low output voltage THD and electromagnetic interference (EMI) are needed. The general function of the multi-level inverter is to synthesise a desired voltage from several levels of DC voltages. For this reason, multi-level inverters can easily provide the high power required of a large EV or HEV drive. As the number of levels increases, the synthesised output waveform has more steps, which produces a staircase wave that approaches the desired waveform (Han et al., 2014).

In case of cascaded H-bridge multi-level inverter, DC sources can be PV array, fuel cells or batteries; one of the foremost attractive feature of the cascaded H-bridge multi-level inverters is that it delivers added number of voltage levels deprived of increasing the DC sources (Lee et al., 2014) and power switches, i.e., asymmetrical cascaded H-bridge multi-level inverter. If the cascaded H-bridge multi-level inverter is designed with equal DC sources, then it is called a symmetrical multi-level inverter or

else an asymmetrical multi-level inverter (Thakre and Mohanty, 2015). In this paper, the asymmetrical CHB MLI is utilised to implement different modulation techniques to generate 31-level output with fewer cells. By using this proposed technique, four DC output voltages are obtained in the ratio of 1 : 2 : 4 : 8 and this technique requires fewer switches, i.e., seven switches and it gives more levels, i.e., 31 levels, and it also gives less THD (Emamalipour et al., 2016). Multi-level inverters are uniquely suited for this application because of the high VA ratings possible with these inverters. Where generated AC voltage is available such as from an alternator or AC generator, a back-to-back diode-clamped converter can convert this source to variable frequency AC voltage for the driven motor. For electric vehicles (EVs), a cascaded H-bridge inverter can be used to drive the traction motor from a set of batteries or fuel cells (Shahir and Babaei, 2016).

Figure 1 Blockdiagram of proposed 31-level inverter (see online version for colours)



Large heavy vehicle applications require considerably larger components and storage units in comparison to the majority of the diesel engines used. The average continuous power range expected is 150–300 kW for the majority of applications, however some applications may be larger than 1 MW. Thus, an expected average or continuous power that exceeds 150 kW will be used to define large HEV applications. The most common applications today are the electric and alternative fuel transit buses. The cyclic operations of the transit bus are ideally suited for EVs and HEVs. The electric transit vehicle has been successful and considerable research and development time has been spent in the electric transit industry, hence, the move to larger hybrid system is inevitable (Gowri Shankar and Belwin Edward, 2017). However, the need for a larger drive system that compares with the performances of presently used diesel buses is a fundamental requirement (Lee et al., 2014).

A solar photovoltaic system can supply electric energy directly to load by converting solar energy into photovoltaic effect. This architecture is flexible and these photovoltaic models are important building blocks. Generally, the other kits are needed to transfer energy to a beneficial form or to store energy for future use. Hence the resulting system will therefore be determined by the energy requires in a specific submission. PV systems can be divided into two different types namely:

- Stand-alone
- Grid-tied (Thakre and Mohanty, 2015).
- *Stand-alone*: These are inaccessible by the distribution of electric grid. The components which are present in this system always depend on which kind of load is being served. The inverter can be removed or exchanged from DC-DC converter only if the DC loads are fed by the PV modules (Chithra and Bharathi Dasan, 2011).
- *Grid-tied*: These can be directly joined with the network of electricity distribution, and this system does not need any battery storage. This system requires lower balance of system components and it removes the necessity of energy storage and costs associated with substituting and recycling batteries for individual clients (Emamalipour et al., 2016).

2 Related work

Suganthi Vinodhini and Samuel Rajesh Babu (2016) suggested a technique of “27 Level/IGBT asymmetrical multi-level inverter using PWM for locomotives”. In this technique, a multi-level inverter is used in case of essentials intended for a great number of electronic switches. This method rises the efficiency of the inverter by reducing the switching losses by using PWM method. This technique uses cascaded H-bridge inverter, and one of the important features of this method is rising the levels and getting an exact sine wave to run the motor easily (Khamooshi and Namadmalan, 2016). In this technique, it reduces the THDs by fast varying the firing pulse for the gates. On the basis of switching techniques and changing the firing angle the new topology generates approximate sinusoidal output voltage and by this the harmonic contents are reduced in this new topology.

Gowri Shankar and Belwin Edward (2016a) introduced a technique of “A 15-level asymmetric cascaded H bridge multi-level inverter with less number of switches for photo voltaic system”. Both symmetrical and asymmetrical multi-level inverters are used in this system. The DC source magnitudes are equal in symmetrical MLI, i.e., $50V_{dc}$, $50V_{dc}$ and $50V_{dc}$, but in case of asymmetrical MLI, the DC source magnitudes are not equal. But both the inverters consist of the same power semiconductor switches only the output voltage levels are dissimilar. The output of symmetrical cascaded H-bridge MLI is nine level, whereas the output of asymmetrical cascaded H-bridge inverter is 15 level. In this technique, THD is less because of using PD-PWM technique (Gowri Shankar and Belwin Edward, 2018). This kind of method is useful in the case of high power applications for photovoltaic systems since it decreases the complete price and size of the system.

Rahman et al. (2017) proposed a technique of “Failure mode analysis for single-phase multi-level qZSI interfacing PV system to utility grid”. Mostly, inverters are affected by different kinds of faults like DC link capacitor fault or switch failure. This failure mode operation is commonly discussed in this paper. In solar panel applications, the quasi Z-source inverter is commonly preferred for its boosting characteristics. Here, multilevel qZSI is implemented for grid synchronisation. The control algorithm for operation of multi-level inverter with fault control (Jonnala et al., 2016). This process is analysed by

using simulation software. From the result, THD is reduced to less than 5%. The current provided to the power grid will be reduced in accord with the power debit. The voltage and current rating of the switching devices used in the QZS multi-level inverter will not be changed due to this process up to three bridge faults (Chithra and Bharathi Dasan, 2011).

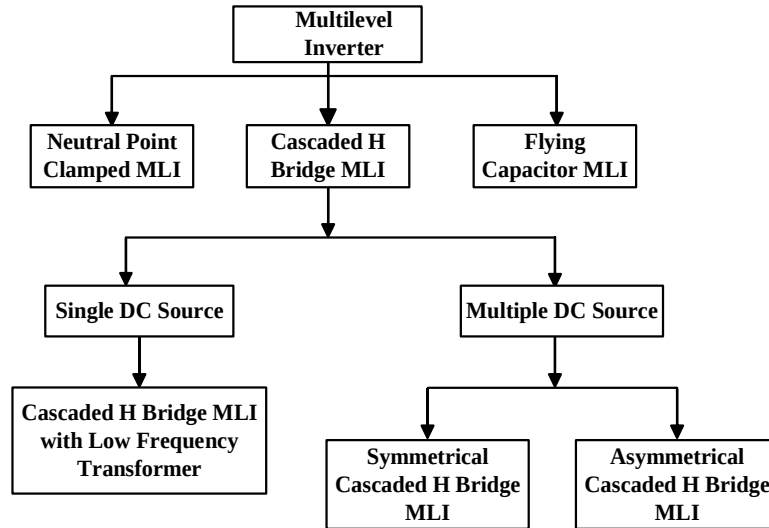
3 Classification of multi-level inverters

Multi-level inverters are divided into three types: neutral point clamped MLI, cascaded H-bridge MLI and flying capacitor MLI.

This cascaded H-bridge MLI is divided into single DC source and multiple DC source. Then multiple DC source is classified into two types:

- symmetrical cascaded H-bridge MLI
- asymmetrical cascaded H-bridge MLI is shown in Figure 2 (Suganthi Vinodhini and Samuel Rajesh Babu, 2016).

Figure 2 Classification of multi-level inverters



3.1 Symmetrical cascaded H-bridge multi-level inverter (SCHBMLI)

In this operation of symmetrical cascaded inverter, each DC sources has a fixed voltage. It generates five levels in output voltage (Fallah et al., 2016). In the exact method, turn on and turn off the switches, then two positive and negative, and zero (0) voltage levels are found. By means of suitable switching of devices, the redundancy in switching state will provide average power control and it is calculated by using (Khamooshi and Namadmalan, 2016),

$$V_{2p} = V_{(2p-1)} = V \quad (1)$$

3.2 Asymmetrical cascaded H-bridge multi-level inverter (ASCHBMLI)

In order to increase the levels of output deprived of raising the cells number (R), asymmetrical nature in DC sources of CHB cells shown in below figure, are selected conferring to a symmetrical development through a factor of 2 or 3. For an R value of cascaded converter cells, it is possible get the below different voltage levels by one phase:

$$N = 2^{R+1} - 1 \quad (2)$$

If $V_{dc,j} = 2^{j-1} V_{dc,j} = 1, 2, 3 \dots R$ (For binary)

$$N = 3^R$$

If $V_{dc,j} = 3^{j-1} V_{dc,j} = 1, 2, 3, \dots R$ (For tertiary)

The determined output voltage of R cascaded cell based MLI is

$$V_{0,Max} = \sum_{j=1}^R V_{dc,j} \quad (3)$$

The above equations can be written as follows

$$V_{0,Max} = (2^R - 1)V_{dc} \quad (4)$$

If $V_{dc,j} = (2^{j-1} - 1) V_{dc,j} = 1, 2, 3, \dots R$ (for binary)

$$V_{0,Max} = \frac{3^R - 1}{2} * V_{dc} \quad (5)$$

If $V_{dc,j} = (3^{j-1} - 1)V_{dc,j} = 1, 2, 3, \dots R$ (for binary)

The above equations show that the asymmetrical multi-level inverter can produce further levels of voltage and developed yield voltage with the same number of cells. In this asymmetrical cascaded H-bridge inverter, the magnitudes of DC sources are not equal. The magnitudes of DC sources are proposed by binary technique of voltage like $25V_{dc}$, $50V_{dc}$, $100V_{dc}$ respectively (Gowri Shankar and Belwin Edward, 2016a). Comparison between symmetrical and asymmetrical MLI is given in Table 1.

Table 1 Comparison between symmetrical and asymmetrical MLI

Description	Symmetrical inverter	Asymmetrical inverter	
		Binary	Tertiary
No. of levels	$2R + 1$	$2R + 1 - 1$	$3R$
No. of DC sources	R	R	R
No. of switches	$4R$	$4R$	$4R$
$V_{0, max}$	R	$2R - 1$	$(3R - 1)/2$

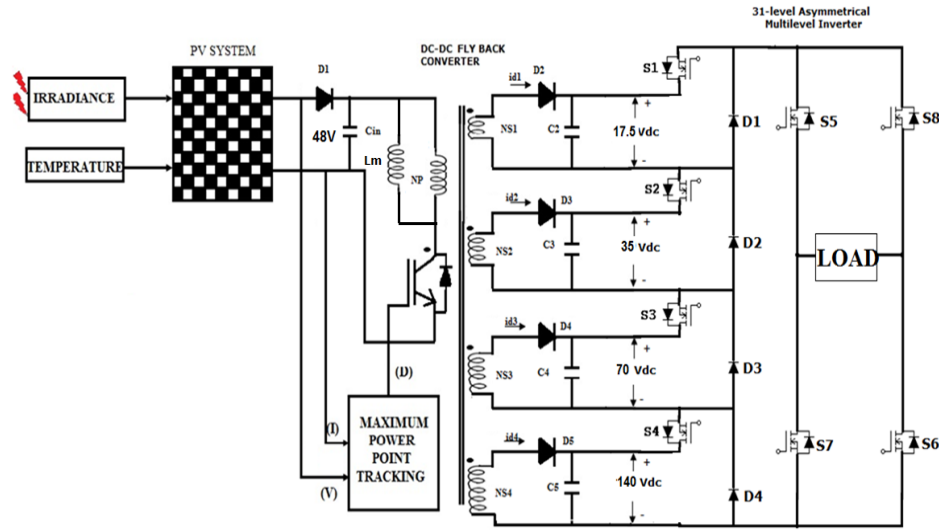
4 Proposed new 31-level MLI topology

In this proposed technique, we are using a single solar panel, the output of this solar panel depends upon the climate conditions, and it gives output of DC voltage, which is variable

or fluctuated. This DC output voltage is given to the multi-output flyback converter, this flyback converter converts the fluctuated DC voltage into regulated voltage. From this, we will take the voltage in the ratio of 1 : 2 : 4 : 8.

This voltage is given to the asymmetrical cascaded H-bridge MLI, here it uses only eight switches in order to get the 31 level. Then it is given to load. Here we are using microcontroller in order to generate the pulses, and we are using opto-coupler for protection purpose, and driver circuit for amplification, it amplifies 5V into MOSFET driving voltage 12V. In this technique, we are using fewer switches and we obtained 31 level MLI, as shown in Figure 3. This proposed method of MLI is asymmetrical standard cascaded H-bridge inverter with fewer switches and thereby escalating the level of the sink (Gowri Shankar and Belwin Edward, 2018).

Figure 3 Circuit diagram of proposed technique (see online version for colours)



4.1 Solar panels

Solar panels are used to generate the electrical energy from sunlight. The solar cell is made by P-N type semiconductor material, which converts solar radiation and temperature into electrical energy. A sum of photovoltaic cells will be associated composed in a 'Module', as shown in Figure 4 and frequently summarised in glass detained a frame and then it can be mounted as essential. In the module, the cells are wired in series or parallel in order to yield an appeased voltage. We will generate 16 volts from sunlight in order to charge the 12V battery.

In maximum situations, association of panels together produce an 'Array'. Similar type of panels can be associated in sequence in order to produce a higher voltage (two 12 V panels are associated in sequence for the production of 24 V). Frequently, in order to produce an increased current, more panels will be connected in parallel. Switching of popular pulse width modulation (PWM) schemes (Baskaran, 2015), which is common in modern power equipment, increases the EMI, and induces internally circulating currents into the motor bearing causing failure and stator winding insulation breakdown of the traction motors (Rahman et al., 2017).

Figure 4 Solar panel (see online version for colours)

4.2 MPPT technique

The electrical power conversion efficiency of the PV panel depends upon the amount of power obtained from solar panel. The electrical power conversion efficiency is mainly based on the sunlight intensity and temperature (Gowri Shankar and Belwin Edward, 2016b). Low electric power conversion may be based on load current, operating temperature and solar-radiation intensity. This would occur as the PV array is connected to a load and the working point would be interconnected (V-I) curves in the solar panel. Receiving a steady power transformation is not easily conceivable in this situation to correct the disadvantage. MPPT is used to improve the power conversion efficiency in solar panel. Here, perturb and observe (P&O) algorithm is used to improve the efficiency of solar panel.

Large heavy electric vehicle applications such as transit buses, recreational vehicles, heavy-duty trucks, school buses, military vehicle, etc., are an additional area that would benefit from a hybrid system. Although some researches have been done in these areas, the overall system implementation becomes more difficult because the vehicles are larger and obviously require a larger ESS, APU, and drive system. In addition, these large components are more expensive and less reliable because the semiconductor switching dV/dt is considerably higher, the devices dissipate more heat due to higher switching losses, and higher rated components tend to cost more (Jonnala et al., 2016).

4.3 Perturb and observe (P&O) technique

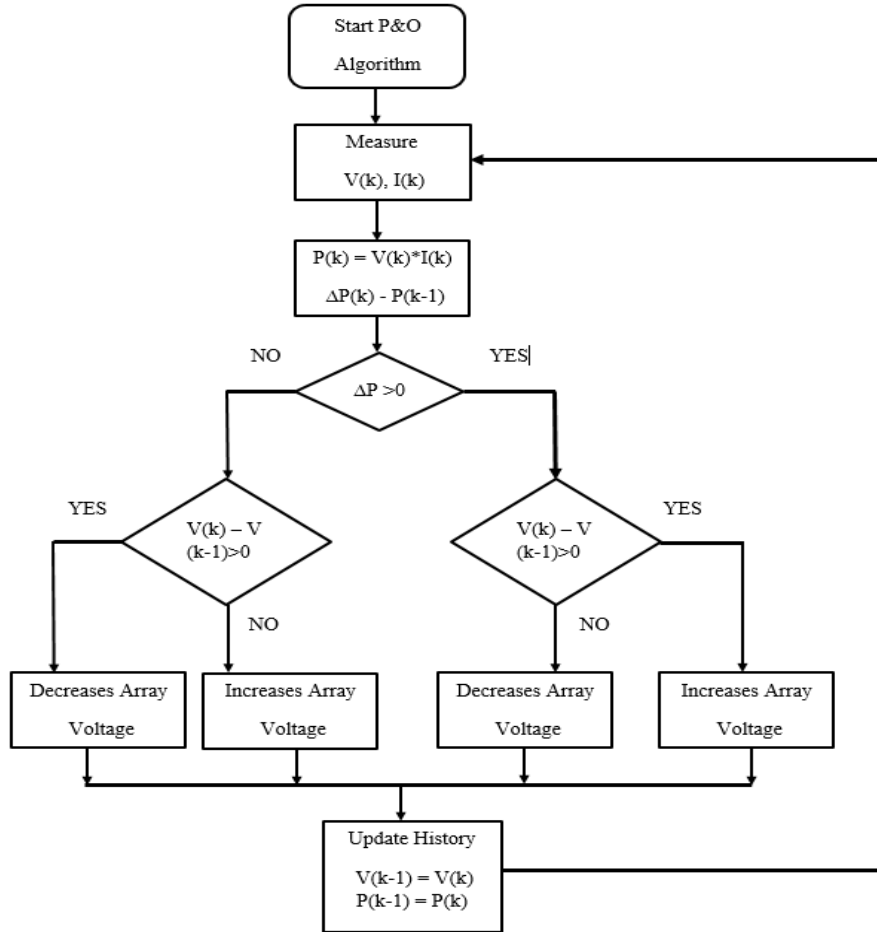
The P&O algorithm and flow chart are given in Figure 5. The advantage we gain from this P&O algorithm is the measured parameters and feedback arrangement to maintain the gate pulse for switch is maximum for obtaining maximum power from solar panels (Fallah et al., 2016).

From the flowchart, the rate of change of power is calculated from current values of voltage and current and previous values. To maintain the variation of power in certain limit we should maintain small perturbation size (Darmawardana et al., 2015).

Let,

P_k : Current power value

P_{k-1} : Previously acquired power value.

Figure 5 P&O flowchart for MPPT

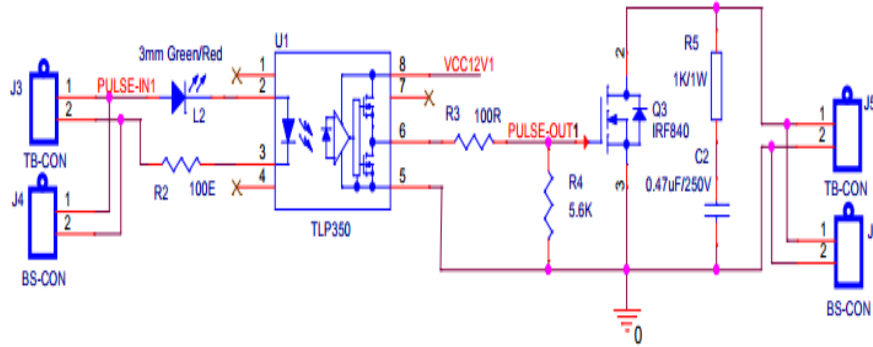
4.4 Opto-coupler and driver circuit for MOSFET

Basically, an opto-coupler is an interface between two circuits that function at dissimilar voltage stages. The important benefit of an opto-coupler is the electrical isolation among the input and output circuits.

The main functions of opto-coupler in the circuit are;

- monitors the high voltage
- output voltage sampling for regulation
- system control micro for power ON/OFF
- ground isolation.

Here TLP 350 IC is used as an opto-coupler with driver IC for driving MOSFET. The circuit used to drive and protect a single MOSFET is given in Figure 6.

Figure 6 Circuit diagram of opto-coupler and driver IC for MOSFET (see online version for colours)

4.5 Cascaded H-bridge multi-level inverter

The H-bridge MLI uses capacitors and switches and needs a reduced number of elements in every stage. This topology comprises sequences of power alteration cells and influence must scale effortlessly. The grouping of capacitor and switches is known as H-bridge (Darmawardana et al., 2015). It provides separate input DC voltage for each H-bridge. It involves H-bridge components and every cell may give three dissimilar voltages correspondingly 0, +ve DC and –ve DC voltages. The main advantage of this kind of MLI is that it requires less quantity of component compared to diode clamped and flying capacitor inverters. And also, the cost and densities are reduced compared to the other two inverters (Baskaran, 2015; Gowri Shankar and Belwin Edward, 2016b).

4.6 Microcontroller

In this proposed technique, we are using the TMS320F2802x piccolo TM microcontroller. The cost of this component is less and development equipment for the piccolo F2802x series in the TI MCU launch pad system, and it is compatible with through several plug-on booster packs. It gives better performance than the others, hence we selected. It includes frequency of 60 MHz C28x CPU, 64 kB Flash, 8 PWM channels, 12-bit ADC, capture interface, UART, SPI, and so on.

5 Modes of operation of proposed technique 31 level MLI

As explained in Figure 6, here we use a single solar panel, the output voltage of this single solar panel is DC voltage and it is given to flyback converter to convert the fluctuated DC voltage into regulated voltage. From this, we will take the voltage in the ratio of 1:2:4:8. This voltage is given to the asymmetrical cascaded H-bridge MLI, here it uses only eight switches in order to get the 31 level. Then it is given to load. The different level of all AC outputs is connected in series to give the voltage waveform, which is sum of the inverter outputs. The output voltage is virtually sinusoidal and therefore the THD is less. The Fourier transform for this topology was carried out: $V(\omega t) = (4V_{dc}/\pi) \sum_n [\cos(n\theta_1) + \cos(n\theta_2) + \dots + \cos(n\theta_s)] * (\sin(n\omega t)/n)$,

where, $n = 1, 3, 5, 7, \dots$. Here the harmonic distortion is reduced compared to the 27-level multi-level inverter. This technique it requires fewer components, also the firing angles for every switching devices are calculated and give reduced THD. Output voltage according to switching on-off condition is given in Table 2.

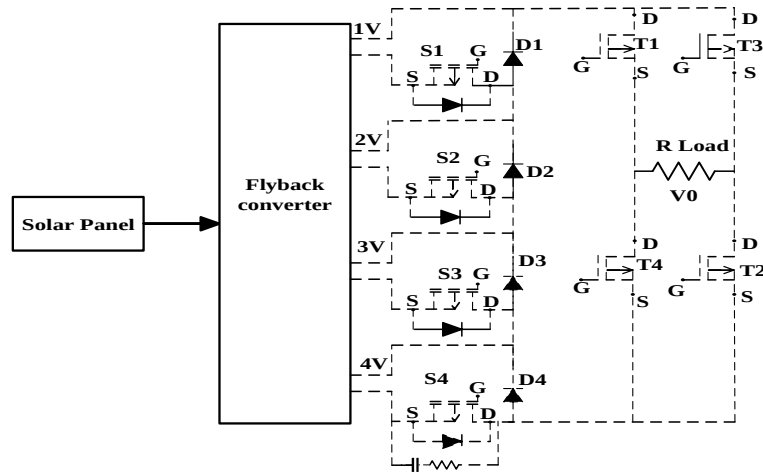
Table 2 Output voltages according to switching on-off condition

V_0 in V_{dc}	S_1	S_2	S_3	S_4	T_1	T_2	T_3	T_4
15v	1	1	1	1	1	1	0	0
14v	0	1	1	1	1	1	0	0
13v	1	0	1	1	1	1	0	0
12v	0	0	1	1	1	1	0	0
11v	1	1	0	1	1	1	0	0
10v	0	1	0	1	1	1	0	0
9v	1	0	0	1	1	1	0	0
8 v	0	0	0	1	1	1	0	0
7 v	1	1	1	0	1	1	0	0
6 v	0	1	1	0	1	1	0	0
5 v	1	0	1	0	1	1	0	0
4 v	0	0	1	0	1	1	0	0
3 v	1	1	0	0	1	1	0	0
2 v	0	1	0	0	1	1	0	0
1 v	1	0	0	0	1	1	0	0
0 v	0	0	0	0	0	0	0	0
-1 v	1	0	0	0	0	0	1	1
-2 v	0	1	0	0	0	0	1	1
-3 v	1	1	0	0	0	0	1	1
-4 v	0	0	1	0	0	0	1	1
-5 v	1	0	1	0	0	0	1	1
-6 v	0	1	1	0	0	0	1	1
-7 v	1	1	1	0	0	0	1	1
-8 v	0	0	0	1	0	0	1	1
-9 v	1	0	0	1	0	0	1	1
-10 v	0	1	0	1	0	0	1	1
-11 v	1	1	0	1	0	0	1	1
-12 v	0	0	1	1	0	0	1	1
-13 v	1	0	1	1	0	0	1	1
-14 v	0	1	1	1	0	0	1	1
-15 v	1	1	1	1	0	0	1	1

*Modes of operation**Mode 0:*

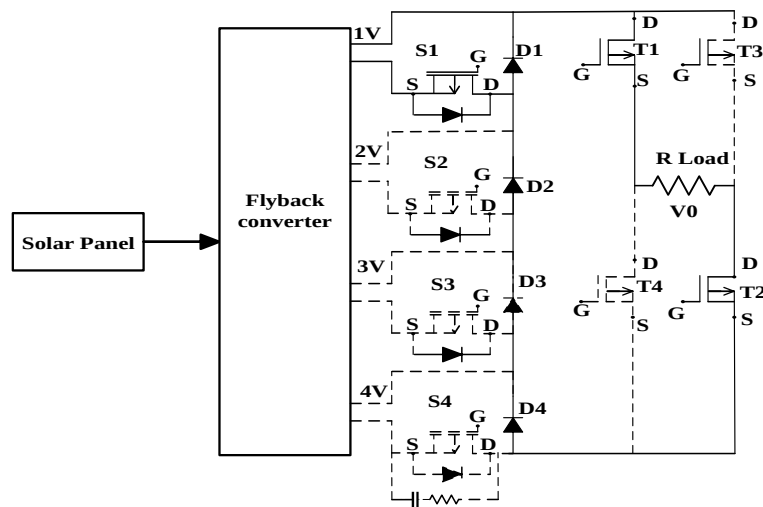
In this mode, the switches S_1 , S_2 , S_3 , S_4 , T_1 and T_2 are turned off. Hence there is no current flow through the circuit, hence we got the output voltage as $0V_{dc}$ as shown in Figure 7 (Gowri Shankar and Belwin Edward, 2016a).

Figure 7 Circuit for Mode 0 operation

*Mode 1:*

In this mode of operation, the switches S_1 , T_1 and T_2 are turned on and the remaining switches S_2 , S_3 , S_4 , T_3 and T_4 are turned off. Hence the current flows through diode D_1 , T_1 , R load, T_2 , D_4 , D_3 , D_2 , D_1 and S_1 and gives the output voltage of $1V_{dc}$ as shown in Figure 8.

Figure 8 Circuit for Mode 1 operation



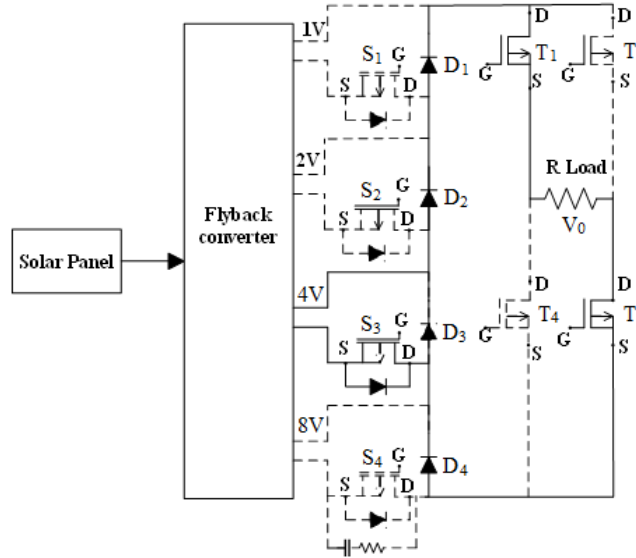
In this operation, S_2 , T_1 and T_2 switches are turned on and the remaining switches S_1 , S_3 , S_4 , T_3 and T_4 are turned off. Here the current flows through D_2 , D_1 , T_1 , R Load, T_2 , D_4 , D_3 and S_2 , hence we got the output voltage of $2V_{dc}$ as shown in Figure 9.

In this mode of operation, the switches S_1 , S_2 , T_1 and T_2 are turned on and the remaining switches S_3 , S_4 , T_3 and T_4 are turned off. In this, current flows through S_1 , T_1 , R Load, T_2 , D_4 , D_3 , and S_2 . Hence, we got the output voltage of $3V_{dc}$ as shown in Figure 10.

Mode 4:

In this mode, the switches S_3 , T_1 and T_2 are turned on, and the remaining switches S_1 , S_2 , S_4 , T_3 and T_4 are turned off. Hence the current flows through D_2 , S_1 , T_1 , R Load, T_2 , D_4 , and S_3 . And the output voltage is $4V_{dc}$ as shown in Figure 11.

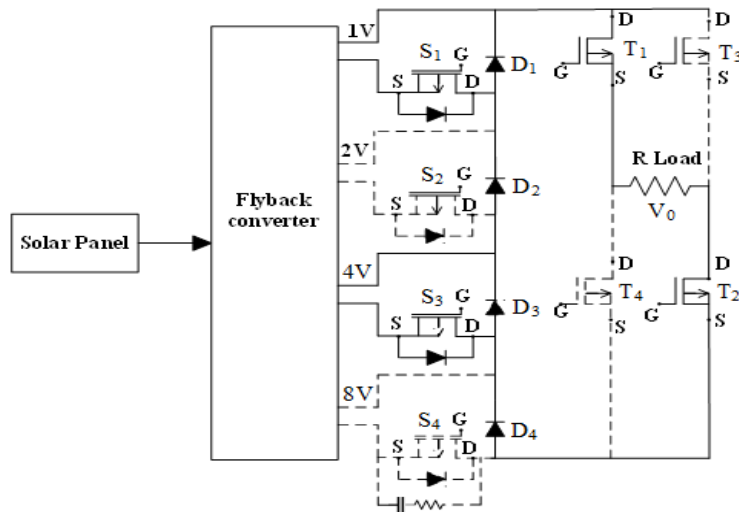
Figure 11 Circuit for Mode 4 operation



Mode 5:

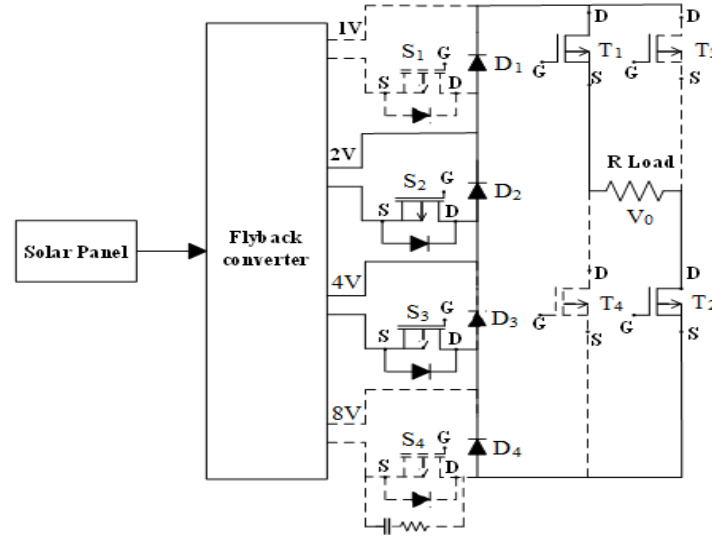
In this mode of operation, the switches S_1 , S_3 , T_1 and T_2 are turned on, and the remaining switches S_2 , S_4 , T_3 and T_4 are turned off. Here the current flows through D_2 , S_1 , T_1 , R Load, T_2 , D_4 and S_3 , and we got the output voltage of $5V_{dc}$ as shown in Figure 12.

Figure 12 Circuit for Mode 5 operation

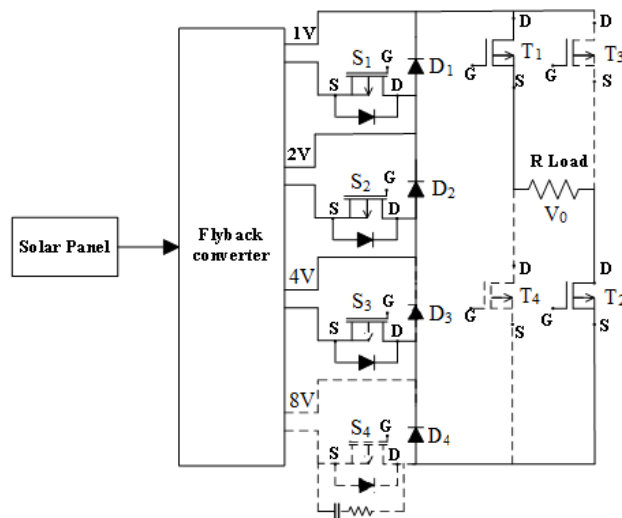


Mode 6:

In this mode, the switches S_2 , S_3 , T_1 and T_2 are turned on and the remaining switches S_1 , S_4 , T_3 and T_4 are turned off. Here the current flows through S_2 , D_1 , T_1 , R Load, T_2 , D_4 , and S_3 . Hence, we got the output voltage of $6V_{dc}$. As shown in Figure 13.

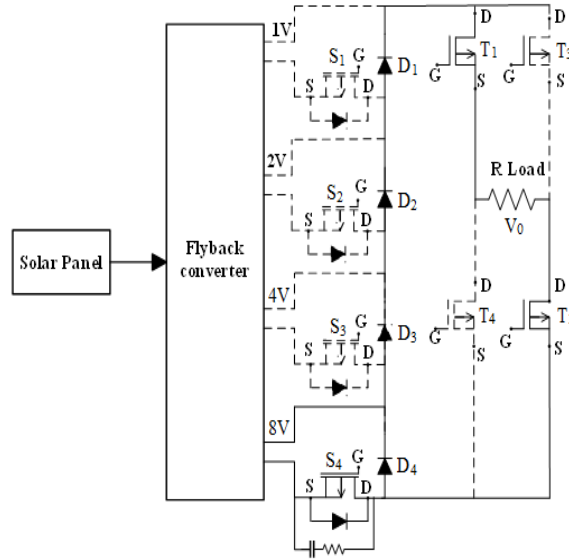
Figure 13 Circuit for Mode 6 operation**Mode 7:**

In this mode, the S_1 , S_2 , S_3 , T_1 and T_2 switches are turned on and the remaining switches S_4 , T_3 and T_4 are turned off. And the current flows through S_2 , S_1 , T_1 , R Load, T_2 , D_4 , and S_3 . Hence, we got the output voltage of $7V_{dc}$ shown in Figure 14.

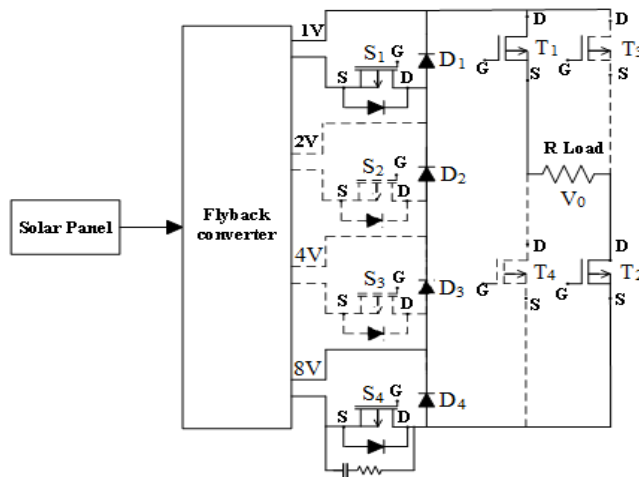
Figure 14 Circuit for Mode 7 operation

Mode 8:

In this mode of operation, the switches S_4 , T_1 and T_2 are turned on and the remaining switches S_1 , S_2 , S_3 , T_3 and T_4 are turned off and the current flows via D_3 , D_2 , D_1 , T_1 , R Load, T_2 , and S_4 . Hence, the output voltage of $8V_{dc}$ as shown in Figure 15.

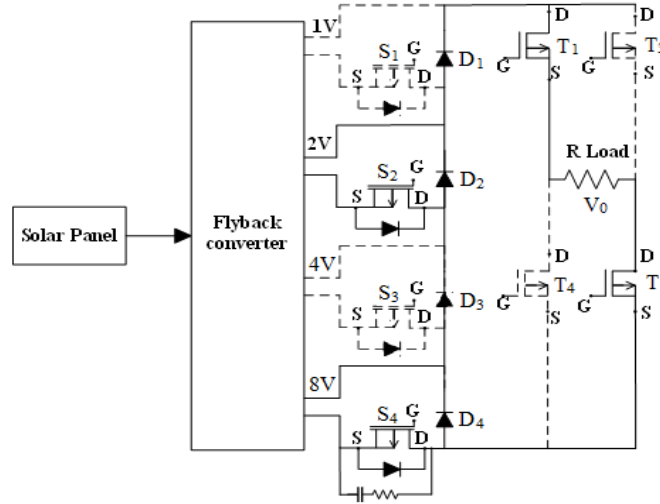
Figure 15 Circuit for Mode 8 operation**Mode 9:**

In this mode of operation, the switches S_1 , S_4 , T_1 and T_2 are turned on and the remaining switches S_2 , S_3 , T_3 and T_4 are turned off. Here the current flows through D_3 , D_2 , D_1 , T_1 , R Load, T_2 , and S_4 . And it produces the output voltage of $9V_{dc}$ as shown in Figure 16.

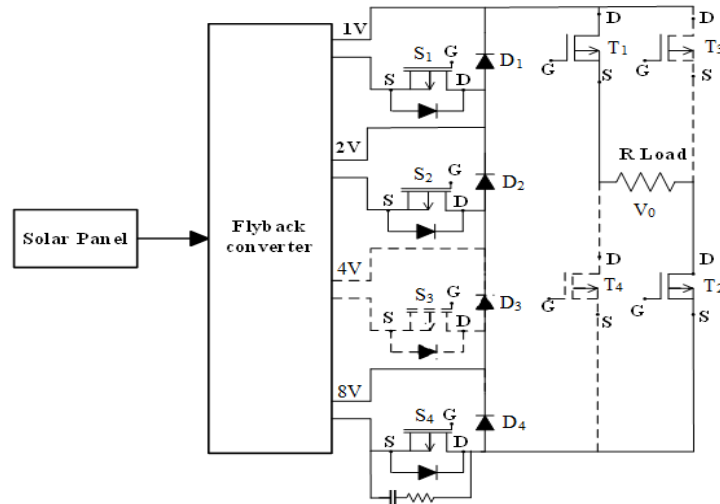
Figure 16 Circuit for Mode 9 operation

Mode 10:

In this mode of operation, the switches S_2 , S_4 , T_1 and T_2 are turned on and the remaining switches S_1 , S_3 , T_3 and T_4 are turned off. And the current flows through D_3 , S_2 , D_1 , T_1 , R Load, T_2 and S_4 . Hence, we got the output voltage of $10V_{dc}$, as shown in Figure 17.

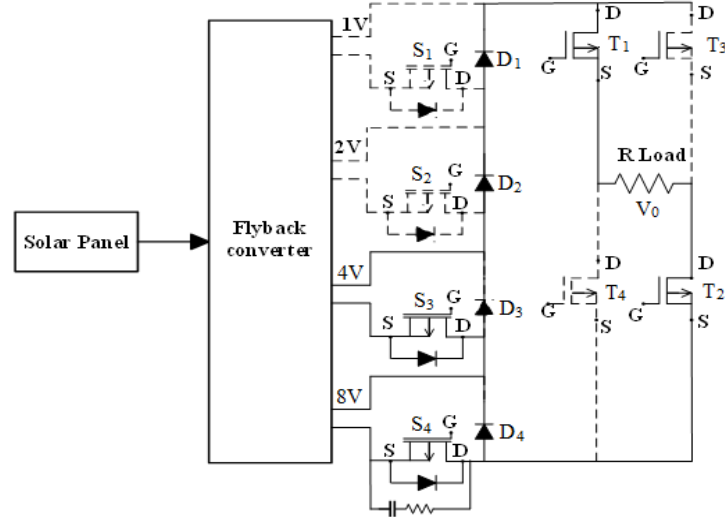
Figure 17 Circuit for Mode 10 operation**Mode 11:**

In this mode of operation, the switches S_1 , S_2 , S_4 , T_1 and T_2 are turned on and the remaining switches S_3 , T_3 and T_4 are turned off. And the current flows through D_3 , S_2 , S_1 , T_1 , R Load, T_2 and S_4 . Hence, we got the output voltage of $11V_{dc}$, shown in Figure 18.

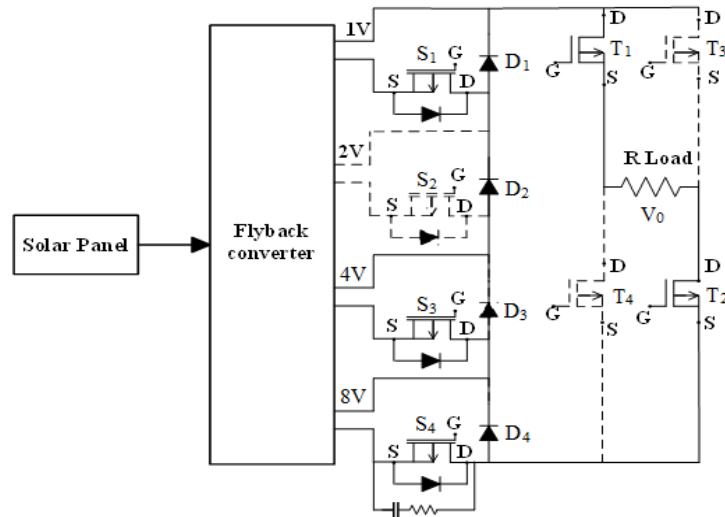
Figure 18 Circuit for Mode 11 operation

Mode 12:

In this mode of operation, the switches S_3 , S_4 , T_1 and T_2 are turned on and the remaining switches S_1 , S_2 , T_3 and T_4 are turned off. And the current flows through S_3 , D_2 , D_1 , T_1 , R Load, T_2 and S_4 . Hence, we got the output voltage of $12V_{dc}$, shown in Figure 19.

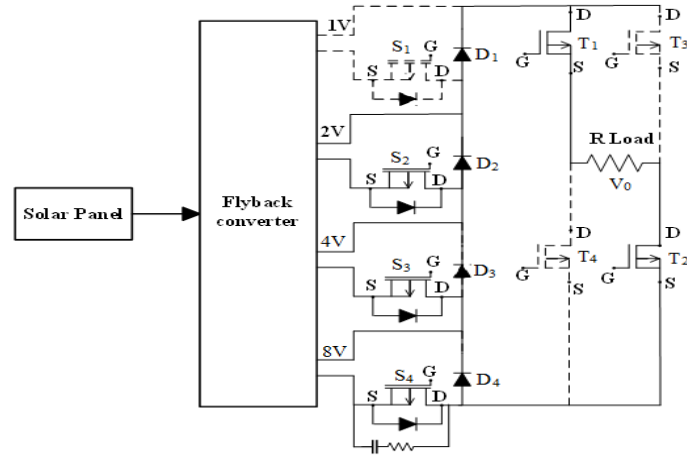
Figure 19 Circuit for Mode 12 operation**Mode 13:**

In this mode of operation, the switches S_1 , S_3 , S_4 , T_1 and T_2 are turned on and the remaining switches S_2 , T_3 and T_4 are turned off. And the current flows through S_3 , D_2 , S_1 , T_1 , R Load, T_2 and S_4 . Hence, we got the output voltage of $13V_{dc}$, as shown in Figure 20.

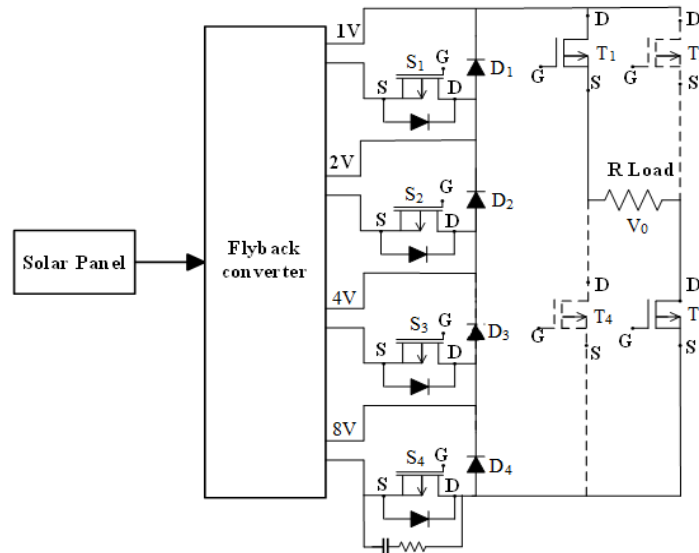
Figure 20 Circuit for Mode 13 operation

Mode 14:

In this mode of operation, the switches S_2 , S_3 , S_4 , T_1 and T_2 are turned on and the remaining switches S_1 , T_3 and T_4 are turned off. And the current flows through S_3 , S_2 , D_1 , T_1 , R Load, T_2 and S_4 . Hence, we got the output voltage of $14V_{dc}$, as shown in Figure 21.

Figure 21 Circuit for Mode 14 operation**Mode 15:**

In this mode of operation, the switches S_1 , S_2 , S_3 , S_4 , T_1 and T_2 are turned on and the remaining switches T_3 and T_4 are turned off. And the current flows through S_3 , S_2 , S_1 , T_1 , R Load, T_2 and S_4 . Hence, we got the output voltage of $15V_{dc}$, as shown in Figure 22 (Gowri Shankar and Belwin Edward, 2016a).

Figure 22 Circuit for Mode 15 operation

6 Simulation circuit

Figure 23 shows the circuit of solar panel with MPPT technique, flyback converter and 31-level MLI, which is implemented in Matlab Simulink.

Figure 24 shows the simulation circuit of flyback converter.

7 Simulation results

Figure 25 shows the output of solar panel, which is given to the flyback converter as input, i.e., this is the input for flyback converter.

Figure 26 shows the DC output voltage of flyback converter. And the output voltage is given to the inverter as a input, i.e., this is the input for inverter.

Figure 27 shows the gate pulses MOSFETs present in multi level inverter.

Figure 28 shows the simulated output voltage for proposed topology.

Figure 29 shows the waveform of 31-level output voltage and THD. From the result, THD present in this result is 3.56%.

8 Hardware results

The hardware setup of 31-level MLI is shown in Figure 30.

Figure 30 shows the hardware circuit of 31 level multi-level inverter. The supply voltage of MLI is given from flyback converter. The output voltage of fly back converter is four different DC voltages at the ratio of 1:2:4:8 depending on the supply voltage. The switching device used in this hardware is MOSFET – IRF840. The gate pulse for this MOSFET switch is generated by using TMS320F2802. This processor is interfaced with MATLAB software to make gate pulse without writing coding. The output pulse amplitude is 3.3V DC. This pulse is unable to drive MOSFET and protect microcontroller from main circuit voltage. Hence, opto-coupler cum driver circuit (TLP350) is used to protect the device and amplify the gate pulse up to 15V DC. Using this pulse, the switches are controlled and make 31-level output voltage across resistive load. This is shown in Figure 30.

Figure 31 shows the hardware circuit for flyback converter. The flyback converter is DC to DC converter, which is commonly used to maintain constant DC voltage for load. Here, flyback converter is used to regulate DC voltage and generate four different kinds of DC voltage as output. The flyback transformer with mult-tapping is shown in Figure 31. These four different voltages of ratio 1 : 2 : 4 : 8 are used as supply voltage of inverter. The switch present in flyback converter is MOSFET-IRF840 and gate pulse for this switch is generated by ATMEGA328. The switching frequency of this switch is 65 kHz. The component details of this circuit is shown in Appendix I.

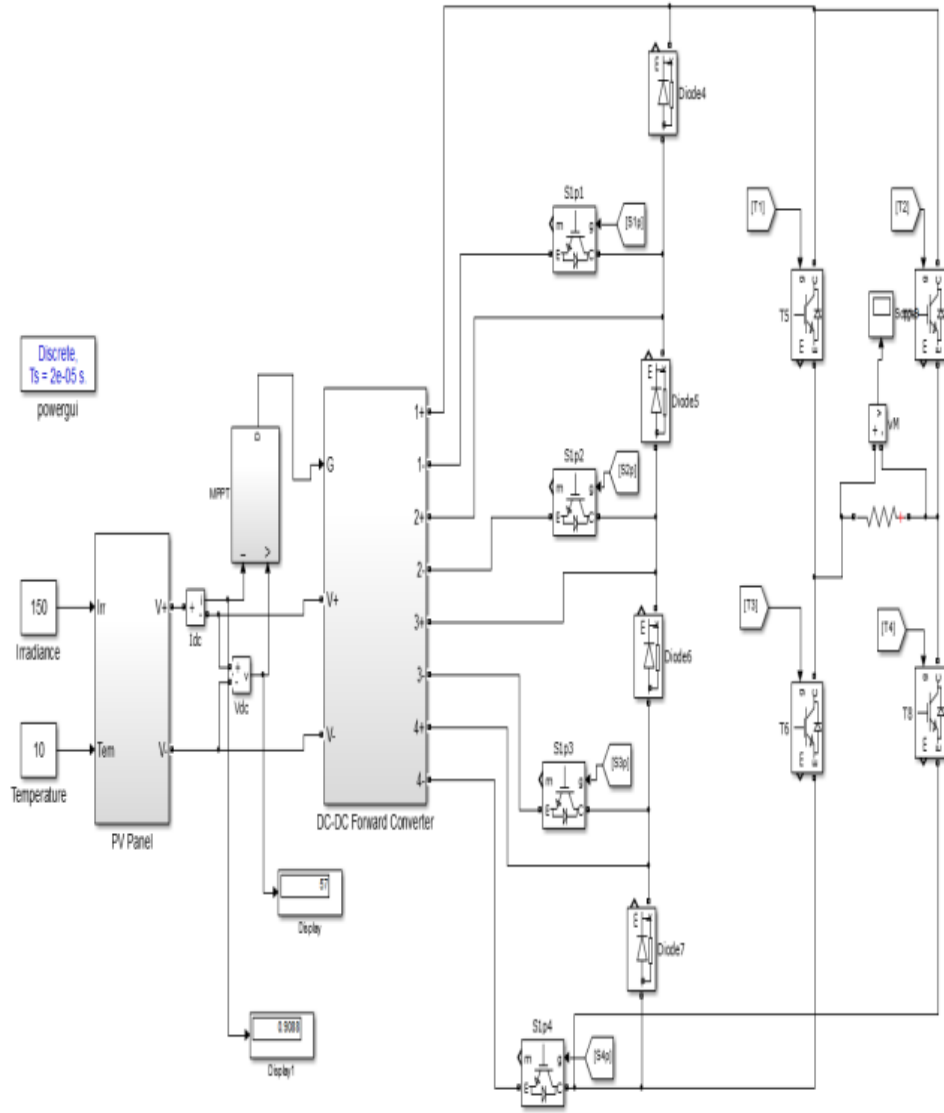
Figure 23 Simulation circuit (see online version for colours)

Figure 32 shows the supply voltage for flyback converter. The $48V_{dc}$ supply is obtained from the solar panel, which is input for flyback converter.

Figure 33 shows the gate pulse for flyback converter (65 kHz).

Figure 34 shows the gate pulse of $6.5V_{dc}$ and $13V_{dc}$ DC voltage from flyback transformer.

Figure 24 Simulation circuit for flyback converter (see online version for colours)

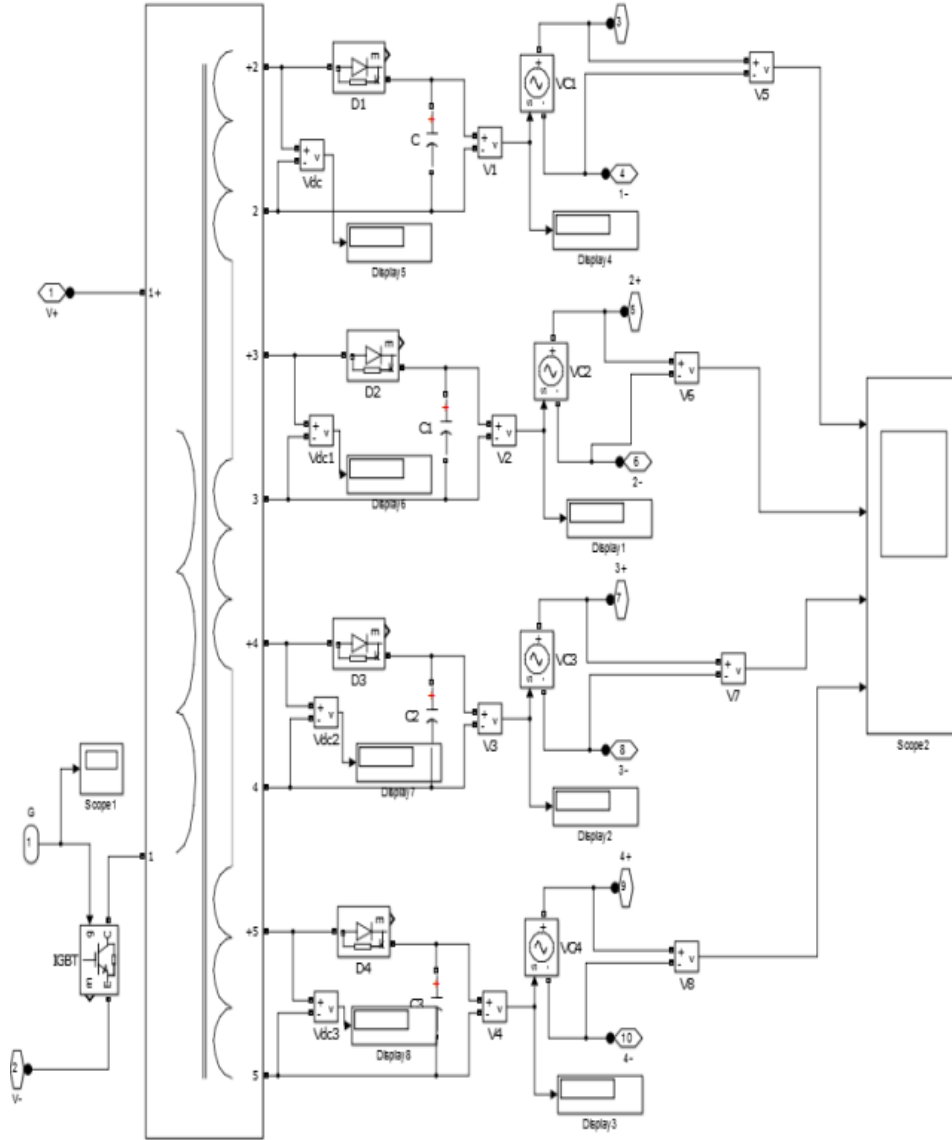


Figure 35 shows the $27V_{dc}$ and $54V_{dc}$ DC voltage from flyback transformer.

Figure 36 shows the gate pulses for S_1 and S_5 switch.

Figure 37 shows the gate pulses for S_2 and S_5 switch.

Figure 38 shows the gate pulses for S_3 and S_5 switch.

Figure 39 shows the gate pulses for S_3 and S_5 switch.

Figure 40 shows the gate pulses for S_5 and S_6 switch.

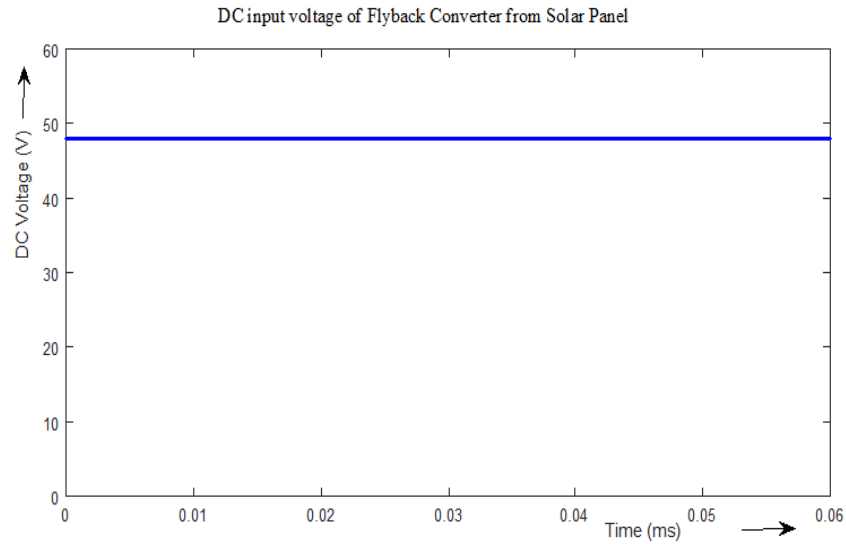
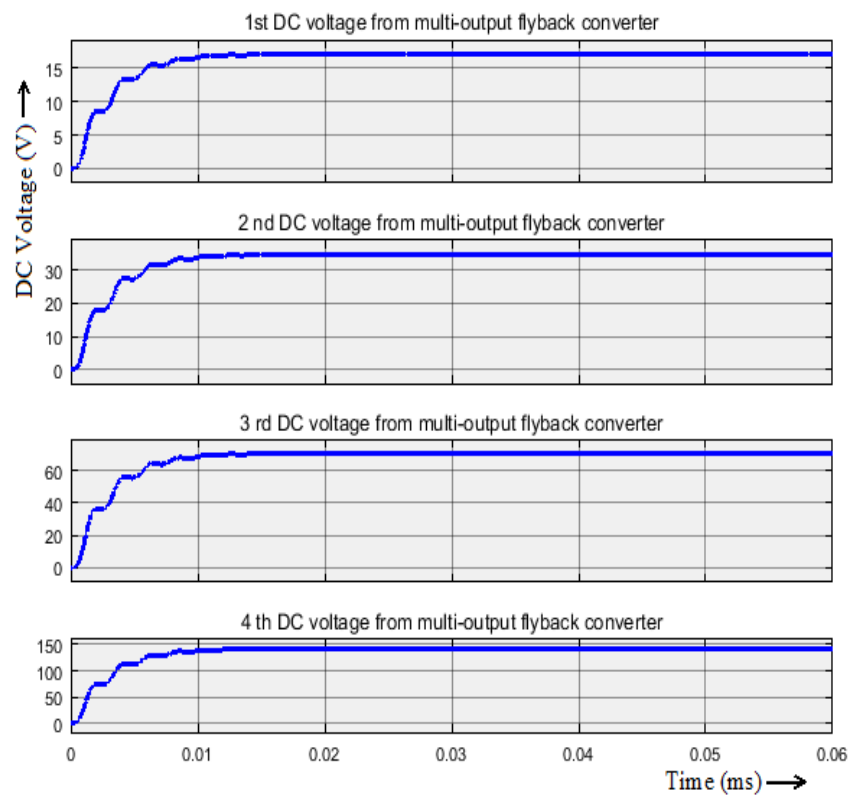
Figure 25 Output voltage from solar panel (see online version for colours)**Figure 26** Output voltage of flyback converter (see online version for colours)

Figure 27 Gate pulses for proposed technique (see online version for colours)

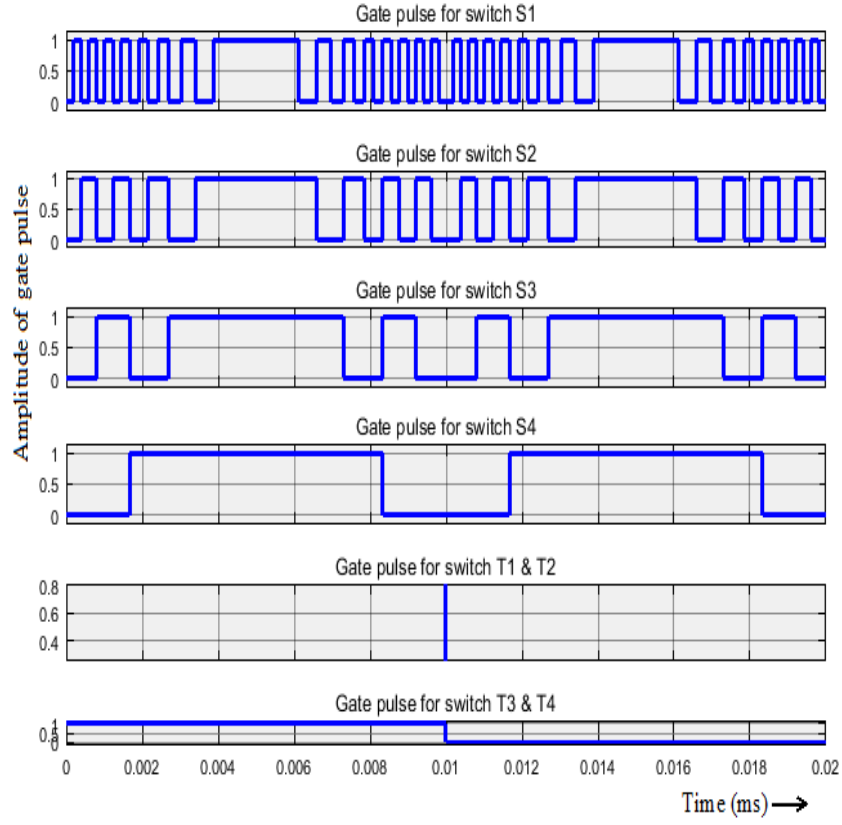


Figure 28 Simulated output voltage for proposed topology (see online version for colours)

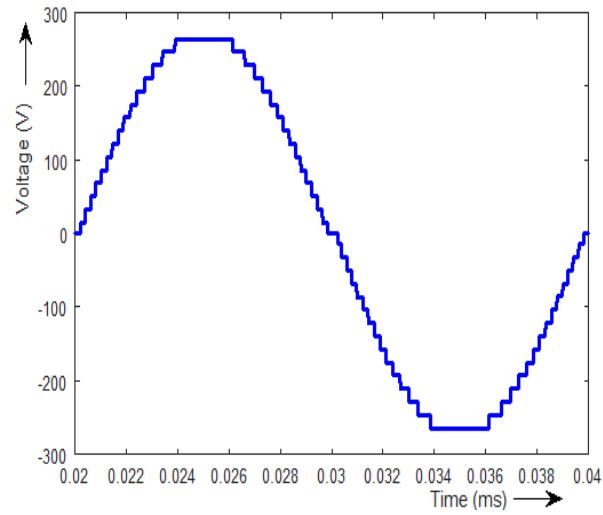


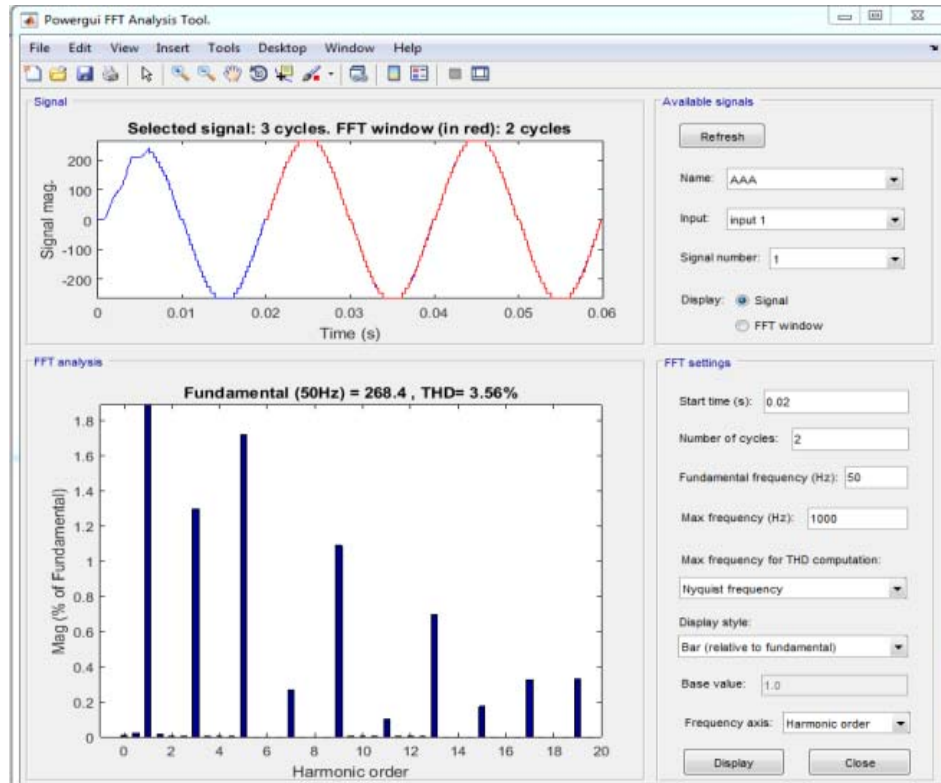
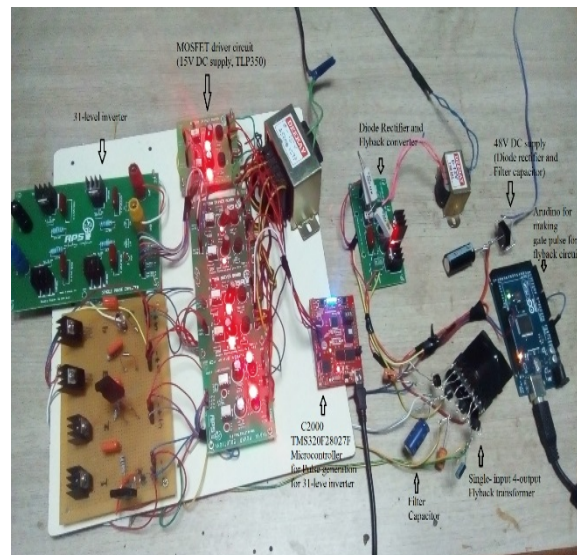
Figure 29 Output waveform of total harmonic distortion (see online version for colours)**Figure 30** Hardware of 31 level multi-level inverter (see online version for colours)

Figure 31 Hardware circuit for flyback converter (see online version for colours)

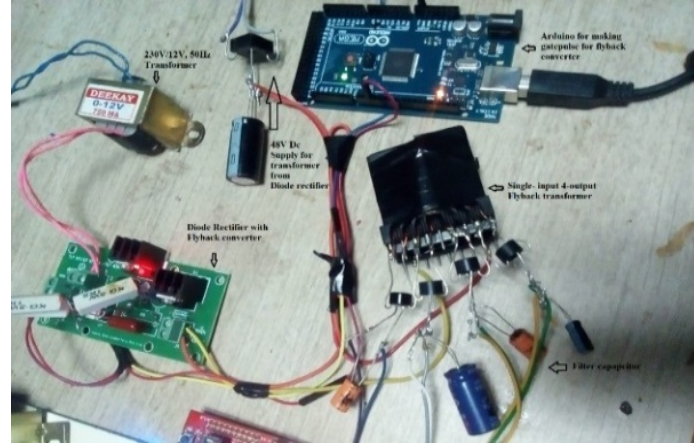


Figure 32 48 DC supply voltage (see online version for colours)

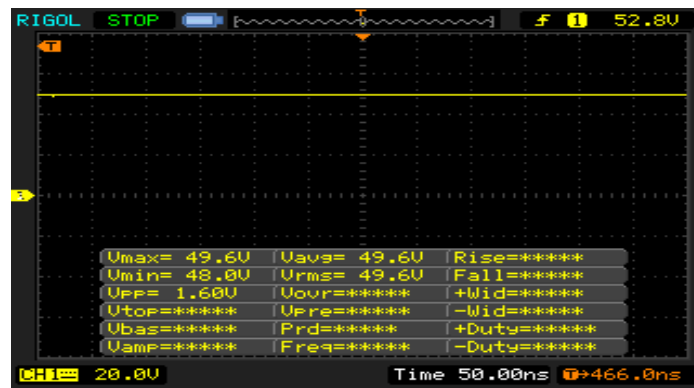


Figure 33 Gate pulse for flyback converter (65KHz) (see online version for colours)

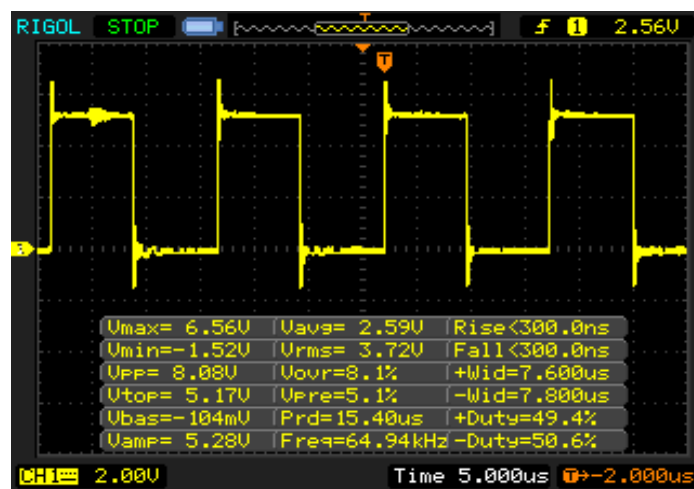


Figure 34 1st and 2nd tapping output voltage from flyback transformer (see online version for colours)

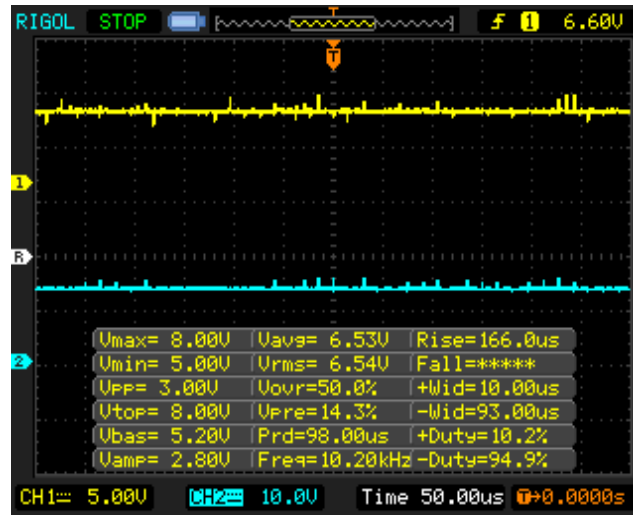


Figure 35 3rd and 4th tapping output voltage from flyback transformer (see online version for colours)

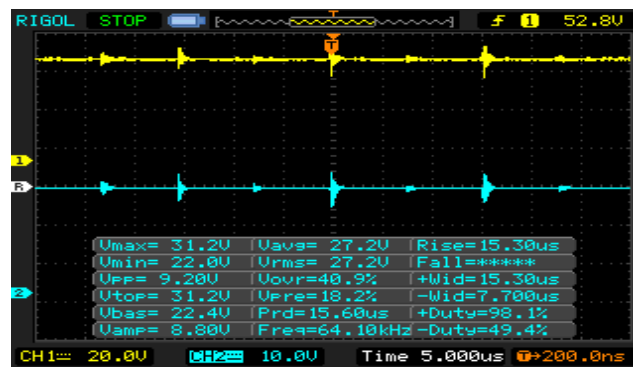


Figure 36 Gate pulses for S_1 and S_5 switches (see online version for colours)

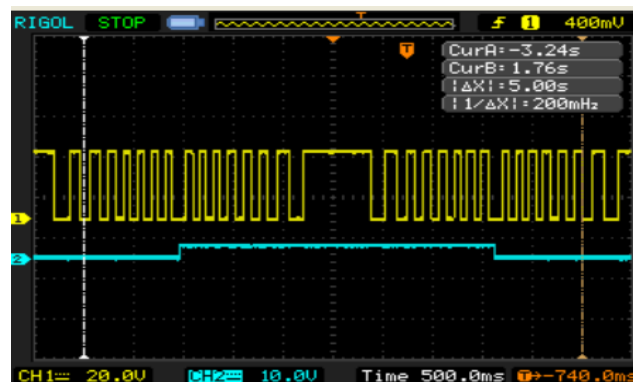


Figure 37 Gate pulses for S_2 and S_5 switches (see online version for colours)

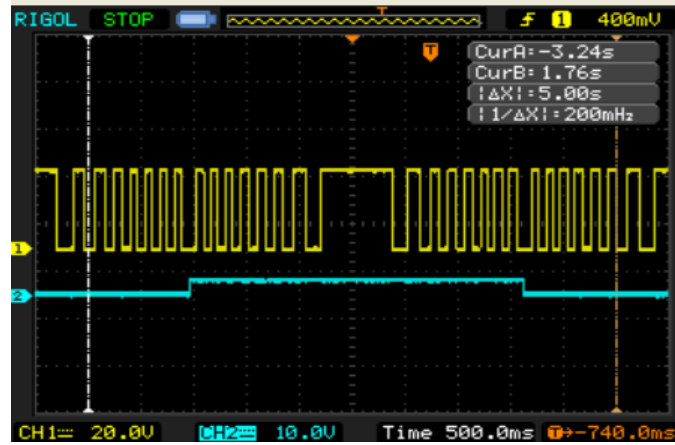


Figure 38 Gate pulses for S_3 and S_5 switches (see online version for colours)

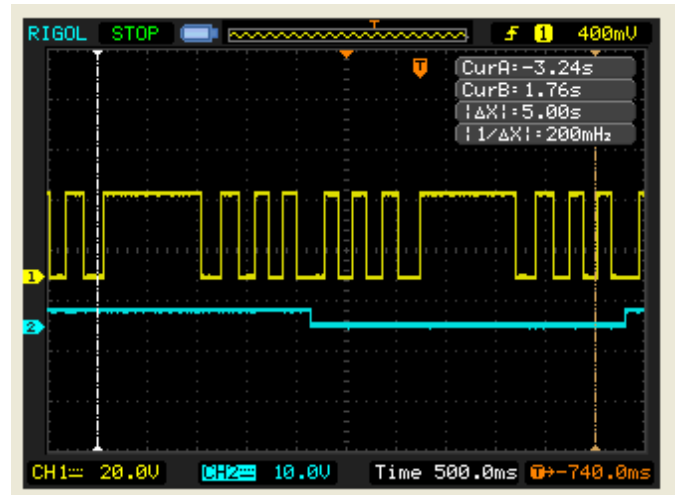


Figure 39 Gate pulses for S_4 and S_5 switches (see online version for colours)

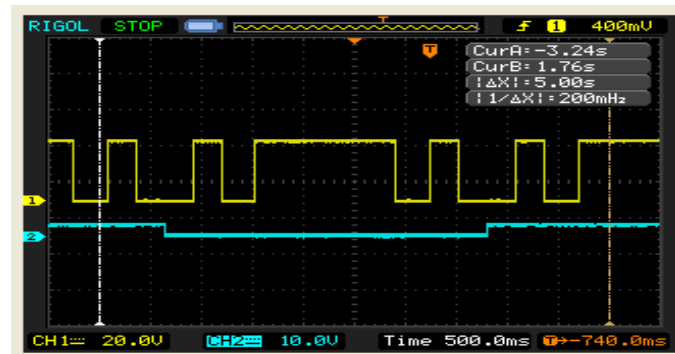


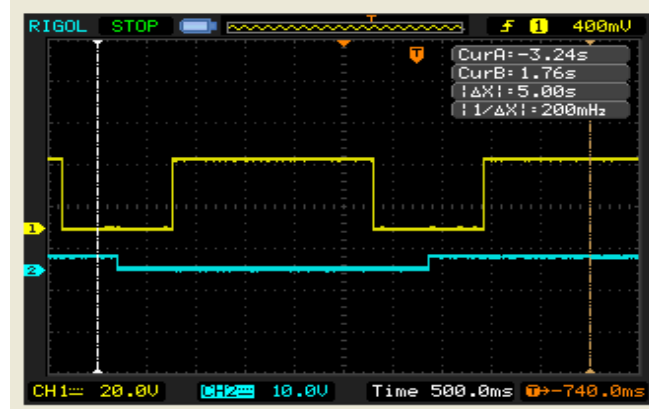
Figure 40 Gate pulses for S_5 and S_6 switches (see online version for colours)

Figure 41 shows the input voltage for H-bridge inverter.

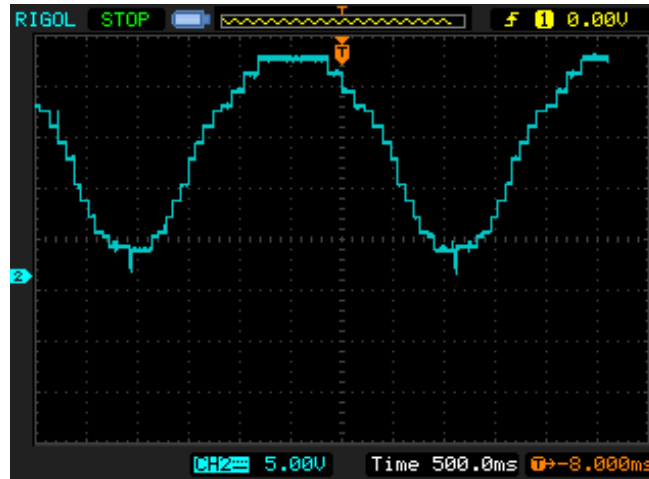
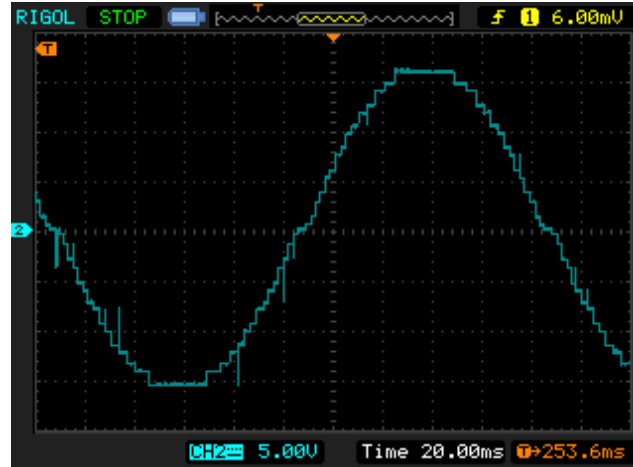
Figure 41 DC input voltage for H-bridge inverter (see online version for colours)

Figure 42 shows the output waveform for proposed 31 level MLI. Figure 30 represents that ASCHBMLI using RL load using TMS320F2802x piccolo TM microcontroller for a real-time system. The input DC voltages are 15V, 30V, 60V and 120V respectively. The output voltage levels are $15V_{dc}$, $30V_{dc}$, $45V_{dc}$, $60V_{dc}$, $75V_{dc}$, $90V_{dc}$, $105V_{dc}$, $120V_{dc}$, $135V_{dc}$, $150V_{dc}$, $165V_{dc}$, $180V_{dc}$, $195V_{dc}$, $210V_{dc}$, $225V_{dc}$, $-15V_{dc}$, $-30V_{dc}$, $-45V_{dc}$, $-60V_{dc}$, $-75V_{dc}$, $-90V_{dc}$, $-105V_{dc}$, $-120V_{dc}$, $-135V_{dc}$, $-150V_{dc}$, $-165V_{dc}$, $-180V_{dc}$, $-195V_{dc}$, $-210V_{dc}$, $-225V_{dc}$. The output voltage waveform of proposed 31 level is shown in Figure 4. And THD analysis of proposed technique is shown in the Figure 27 for ASCHBMLI by using RL load.

Figure 42 Output waveform of proposed technique 31 level MLI (see online version for colours)

9 Conclusion

This paper offers an innovative technique for asymmetrical cascaded H-bridge inverter. The foremost benefit of this proposed method is here. This technique requires few switches and by using fewer switches more levels are obtained, thereby getting a proper sine wave to run the AC motor smoothly. In this we are taking output in the ratio of 1 : 2 : 4 : 8. And the output voltage is virtually sinusoidal and therefore the THD is less. The THD is much less in ASCHMLI and it is implemented by using the TMS320F2802x piccolo TM microcontroller. This kind of proposed technique is helpful in the field of power application in photovoltaics and it reduces overall cost as well as size of the system.

10 Future work

For all traction persistence MLI being an expertise for standard voltage and high-power submissions, in all MLI, only estimated sinusoidal output with harmonic distortion is produced. Then this can be overcome, by decreasing the number of switches, by altering the semiconductor switches and also by correct switching methods like SVM, SPVM, selective harmonic injection method, etc. In all predictable inverters EMI, EMC and switching losses can be diminished by using the above declaration in future MLI.

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Appendix I: Components list

S. No.	Components	Range	Type
Solar panel			
1	Voltage	48V	DC
	Current	6.3A	
	Power	200W	
Flyback converter			
2	MOSFET-IRF840	500V,8A	AC
3	f _{sw}	65kHz	
4	Transformer primary turns	24	
5	Secondary side turns	3,6,12,24	
6	Filter capacitor	100μF, 16V	
		47μF, 63V	
		100μF, 160V	
Multi-level inverter			
7	MOSFET-IRF840	500V, 8A	
8	Diode	MUR3060, 1N2007	
9	Frequency	50Hz	
Gate pulse generation			
10	TMS320F2802-processor	3.3V	
11	ATMEGA 328	3.3V	
Opto-coupler and driver circuit			
12	TLP 350	5V to 15V	DC
13	Supply voltage	15V	AC
14	Bridge rectifier-W10		
15	Capacitor	470μF,25V	